Modern Academy
for Engineering and Technology
Basic Science Department

Academic Year: 2019-2020
Semester: Fall
Exam Date: 29/12/2019

QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject: (MTH203) Differential Equations and Transforms **Spec:** Comp. and Comm.
Examiners: Assoc. Prof. Dr. Ashraf Taha and Prof. Dr. Aly Essawi **Time:** 3 hours

Number of Pages: 2 Number of Questions: 5 Attempt all questions

Question 1 (14 points)

Solve the following differential equations:

a) $\frac{dy}{dx} = e^x \sec(y) + \sec(y).$ (3 points)

b) $\frac{dy}{dx} = -\sin(x-y) + 1.$ (4 points)

c) $\left(3x^2 + 2y\sin(2x)\right)dx + \left(2\sin^2(x) + 3y^2\right)dy = 0.$ (4 points)

d) $\left(1+x^2\right)\frac{dy}{dx} - 2xy = x\ y^3.$ (3 points)

Question 2 (16 points)

Find the general solution of the following differential equations:

a) $\frac{d^5y}{dx^5} - 3\frac{d^4y}{dx^4} + 6\frac{d^3y}{dx^3} - 4\frac{d^2y}{dx^2} = 0.$ (4 points)

b) $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 2\sin(x).$ (4 points)

c) $\frac{d^2y}{dx^2} + y = \tan(x)\sec(x).$ (4 points)

d) $\frac{d^2y}{dx^2} + y = 1,$ given that $y = \cos(x)$ is a homogenous solution. (4 points)

Question 3 (16 points)

a) ***Solve the following differential equation using Laplace transform:***

$\frac{d^2y}{dt^2} - 3\frac{dy}{dt} + 2y = 4; \qquad y(0) = 1, y'(0) = 0.$ (4 points)

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b) ***Find the Laplace transform of the following functions:***

i. $f(t)=\cos\left(3t+\frac{\pi}{2}\right)+\cosh t.$ (3 points)

ii. $f(t)=u_5(t)\ e^{t-5}\ t^2.$ (3 points)

iii. $f(t)=t\left(\sin t+e^{2t}\right).$ (3 points)

iv. $f(t)=\sin 2t\ \cos 2t+\int_0^t\left(\cosh\tau+1\right)\ d\tau.$ (3 points)

Question 4 (16 points)

a) ***Find the inverse Laplace transform of the following functions:***

i. $F(s)=\frac{s^3-s^2+s-1}{s^5}+\frac{4s}{s^2+16}.$ (3 points)

ii. $F(s)=\frac{e^{-s}-e^{-2s}}{s^2-1}.$ (3 points)

iii. $F(s)=\frac{1}{s\left(s^2+1\right)}.$ (3 points)

iv. $F(s)=\ln\left(\frac{s+6}{s+2}\right).$ (3 points)

b) ***Represent the following function*** $f(x)=x$, $0\le x\le\pi$ ***as Fourier Cosine Series*** (4 points)

Question 5 (8 points)

a) ***Prove the recurrence relations*** $xP_n(x)=\frac{n+1}{2n+1}P_{n+1}(x)+\frac{n}{2n+1}P_{n-1}(x).$ (3 points)

b) ***Evaluate the following integrals:***

i. $\int_{-1}^{1} P_1(x)P_2(x)\ dx.$ (1 point)

ii. $\int_{-1}^{1} P_0(x)P_3^2(x)\ dx.$ (2 points)

iii. $\int_{-1}^{1} x\,P_7(x)P_8(x)\ dx.$ (***Hint: use above Prove***) (2 points)

GOOD LUCK